

Evaluation of Machine Learning Models to Forecast Inflation: Bangladesh as a Case Study

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Abstract—In Bangladesh, inflation is a paramount factor that significantly impacts governmental and non-governmental policies. It intricately links the financial value of money and the economy's stability. The decline of a fiat currency's purchasing power over time is referred to as inflation. It is essential to the creation of global central banks' macroeconomic policies. When growth rates are high, inflation increases, and deflation occurs when they are low or negative. We conducted a study to determine the relationship between inflation and the CPI (Consumer Price Index). To determine the historical pattern of inflation, we have put up some approaches based on machine learning to help us predict inflation. Despite the constraints caused by the lack of pertinent data, we conducted a thorough analysis using various machine learning models, including Support Vector Regression (SVR), ARIMA, LSTM, Prophet, XGBoost, and GRU. Each model was rigorously evaluated with different metrics to find the accuracy. This research clarifies the intricate connection between inflation and the CPI and emphasizes the crucial role that data accessibility plays in the development of predictive models. Our findings will encourage further research in this area, resulting in future economic forecasts that are more precise and insightful.

Index Terms—Inflation, LSTM, ARIMA, Time-Series Forecasting, XGboost Regression, GRU, AutoML, Prophet, SVR, SARIMA

I. INTRODUCTION

Inflation forecasting is crucial in formulating effective monetary policy, especially in economies where rational agents rely heavily on inflation predictions for economic commitments, wages, and interest rates [14]. Developing accurate inflation projections are essential, particularly in countries like Bangladesh, which experiences substantial and unpredictable inflation. The emergence of machine learning (ML) and big data techniques in the literature on inflation forecasting has brought about advancements in prediction accuracy compared to traditional models like the Phillips curve [15]. However, despite the progress made, there still needs to be a universally

agreed-upon method for evaluating the performance of inflation forecasts. Accurate inflation predictions are paramount for implementing effective monetary policies and managing return on investment (ROI) [18]. The analysis focuses on univariate time series modeling, with specific attention given to popular models such as autoregressive integrated moving averages (ARIMA), long short-term memory (LSTM) networks, Support Vector Regression (SVR), Prophet from Facebook, AutoML for XGBoost, SARIMA (Seasonal ARIMA) and Gated Recurrent Unit (GRU). ML models have demonstrated superior outcome accuracy in predicting inflation compared to traditional econometric models [16] [17]. By addressing the existing gap in the literature on inflation forecasting in emerging economies like Bangladesh, this research aims to identify the most effective model for predicting inflation through appropriate diagnostic and assessment criteria. We will look at our study's methodology, data analysis, model assessment, and results in the following parts. Moreover, it seeks to generate precise predictions of consumer price inflation in Bangladesh across various time horizons. Moreover, the findings of this research have the potential to extend beyond inflation forecasting and offer valuable insights into forecasting other econometric variables and financial solvency.

II. LITERATURE REVIEW

This research aims to predict inflation using comparisons between univariate and multivariate models. As a result, the literature review mainly concentrates on the critical works on inflation forecasting. The Box-Jenkins ARIMA technique was employed for modeling inflation in most studies conducted worldwide. Iftikhar et al. [1] in Pakistan; Okafor et al [2]; John et al. [3]; Mustapha et al. [4]; Popoola et al. [5] in Nigeria; Jere et al. [6] in Zambia; Islam et al. [7] and Habibah et al. [8] in Bangladesh found ARIMA model to be the better model for forecasting inflation. Some studies identified the seasonal autoregressive integrated moving average (SARIMA) model

as the best ideal by enhancing the seasonal component. For instance, SARIMA was utilized for modeling and forecasting inflation by Akhter et al. [9] in Bangladesh, Out et al. [10] in Nigeria, and Lidiema et al. [11] in Kenya. Akhter et al. [9] used the monthly consumer price index (CPI) from January 2000 to December 2012 in conjunction with the seasonal autoregressive integrated moving average (SARIMA) model to predict Bangladesh's short-term inflation rate. Despite recent attempts by Islam et al. [7] and Habibah et al. [8] to forecast inflation using the ARIMA (1, 0, 0) and ARIMA (3, 0, 0) models, respectively, their predictions fell short of reality as we discovered that the actual inflation rate obtained from the Bangladesh Bureau of Statistics (BBS) varied significantly in the years that followed. So, Bangladesh needs a comparable model study to predict the inflation rate.

III. METHODOLOGY

A. Dataset

The Bureau of Statistics (BBS) of the Central Bank of Bangladesh used the CPI method to compile monthly inflation rates for Bangladesh, with 1995–1996 serving as the base year. From January 2013 to October 2022, the data were forecasted, and the remaining data were preserved as out-of-sample to evaluate the precision of our prediction. To forecast the econometric parameters, 118 data points from January 2013 to October 2022 have been considered.

B. Empirical Models

1) *Long Short-Term Memory (LSTM)*: Recurrent neural networks (RNNs), like LSTM, are one type of RNN that may recall values from prior stages to use for forecasting in the future [7]. RNNs primarily use sequential series to understand the past and forecast the future. The LSTM model was chosen because it can also reflect long-term interdependence between the variables typically found in macroeconomic contexts. Long-Short Term Memory (LSTM) was introduced by Habibah et al. [8] to address the problem of vanishing and exploding gradients by offering a novel architectural approach to RNNs. The following equations describe the behavior of the LSTM model [13]:

$$\begin{aligned} f_t &= \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f) \\ i_t &= \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \\ c_t &= \tanh(W_{hc}h_{t-1} + W_{xc}x_t + b_c) \\ c_t &= f_t c_{t-1} + i_t c_t \\ o_t &= \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \\ h_t &= o_t \tanh(c_t) \end{aligned}$$

Where f_t , i_t , and o_t are the activation of the input, forget, and output gates at time-step t , which control how much of the input and the previous state will be considered, and how much of the cell state will be included in the hidden activation of the network. The protected cell activation at time step t is denoted by c_t , whereas h_t is the activation given to the next layer. The notations of the weight matrices $W_{(.,.)}$ are explicitly denoted with subscripts of their transform. $b_{(.)}$ denotes the bias vector of each gate. W_{xf} and W_{hf} are the matrices that transform the input x_t and hidden state h_{t-1} , respectively, to

the forget gate dimension, W_{xi} and W_{hi} transform the input and hidden state to the input gate dimension and so on. There is simplified notation combining the input and hidden state into a single matrix, thus having a single matrix per gate W_f , W_i , W_c , W_o .

- *Feature Selection with Granger Causality Test*: We take the initial differences for each feature and check for stationarity using the same ADF test procedure to evaluate which characteristics help forecast core CPI.
- *Splitting Multivariate Sequences into Samples with 12 Step In and 1 Step Out*: Here, we will keep the train set from January 2013 to November 2019 and the test set from December 2019 to October 2022.
- *Model Training*: A GridSearchCV was made to determine the ideal batch size and epoch of 128 and 250 epochs, respectively, with one verbose while concurrently stacking an additional LSTM layer. EarlyStopping and Dropout regularization approaches were used because the model has severe overfitting issues. The hyperparameters were adjusted manually throughout the experimental runs, and the outcomes were carefully assessed.

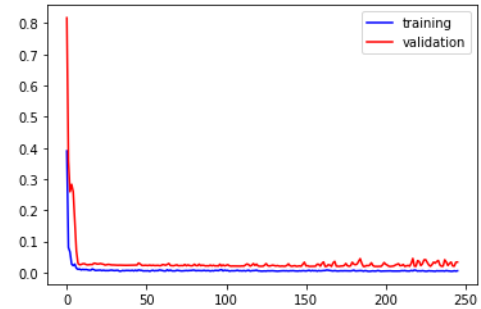


Fig. 1. Training Vs. Validation

- *Feature Importance*: Using the perturbation effect, we measured and determined the feature relevance that affected the recurrent neural network, and Core CPI was the most crucial.

2) *Gated Recurrent Unit (GRU)*: The GRU (Gated Recurrent Unit) model used in this research is a type of recurrent neural network (RNN) that is particularly effective in capturing temporal patterns and dependencies in sequential data. Here, the GRU layer has 64 units and expects input sequences of length with one feature (inflation value) at each time step. After the GRU layer, a Dense layer with one unit is added to produce the final output prediction. The model is compiled using the Adam optimizer. The model is trained for 250 epochs with a batch size of 128. The verbose argument is set to 1 to display the training progress. Overall, the GRU model in this code learns the patterns and dependencies in the historical inflation data during the training phase and uses that knowledge to make predictions on unseen data.

3) *Autoregressive Integrated Moving Average (ARIMA)*: The AR stands for Autoregression, the I for Integrated stands for the differencing of raw observations to enable the time

series to become stationary, and the MA stands for Moving Average, which incorporates the dependency between an observation and a residual error from a moving average model applied to lagged observations. ARIMA employs lagged moving averages to smooth time series data and time series data to forecast future values based on historical values.

- *Seasonal Decompose*: The "Seasonal" graph reveals a clear cyclical tendency that suggests seasonality in the dataset.

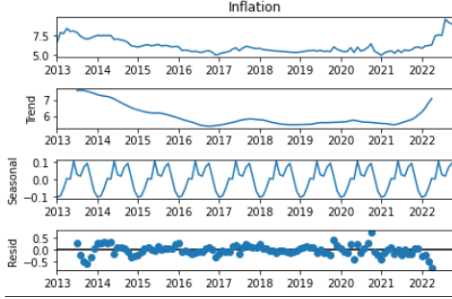


Fig. 2. Seasonal decompose

- *Augmented Dickey-Fuller Test*: The dataset's p-value of 0.676416 after doing the Augmented Dickey-Fuller Test shows that it does not support the null hypothesis; for the data to do so, the p-value must be greater than 0.05. Additionally, this indicates that the dataset is not stationary. The data must, therefore, be eliminated because it is non-stationary and has unit roots and seasonality.
- *Removing Seasonality with Differencing*: We do first- and second-order differencing on the inflation values to keep the data consistent. Next, we plot the ACF and PACF plots to see if the stationary property has changed. The

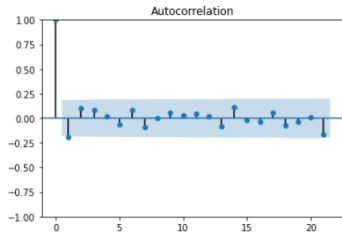


Fig. 3. ACF plot

ACF plot reveals that the series has decreased in seasonality significantly from the first differencing, indicating that the time series is stationary.

- *Finding best parameters*: A loop was created to iterate by different combinations of p,d, and q values. It sought to find the combination for a model that would give the lowest RMSE score. The best combination was the order of (5,0,6) with an RMSE of 0.986, so we trained our model along these parameters.

4) *Support Vector Regression (SVR)*: SVR is a variation of Support Vector Machines (SVM) used for regression tasks. It finds a hyperplane that maximizes the margin while still

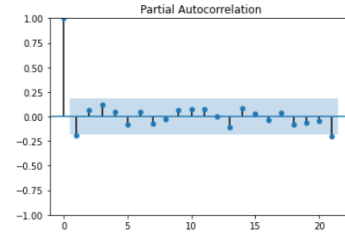


Fig. 4. PACF plot

fitting the data within a specified tolerance. Feature scaling, like standardization and normalization, often benefits SVR when the input characteristics are scaled to a similar range. This makes sure that no single feature takes control of the optimization process. Unlike linear regression models, SVR models are not automatically interpreted. However, the support vectors and their corresponding weights might show how the model makes decisions and the significance of various features when making predictions. [19].

5) *Automated Machine Learning(AutoML)*: The H2O AutoML interface is made to have as few parameters as possible, requiring the user only to supply the dataset most of the time. It automates creating, testing, and comparing many models using various hyperparameters and techniques. In this case, the H2O AutoML model incorporates the XGBoost (Extreme Gradient Boosting), an ensemble learning algorithm known for its performance and scalability. It builds a robust predictive model by combining multiple weak models (decision trees) iteratively [20]. From regression metrics, we determine that the "Date" variable has the highest relative importance, followed by the "CPI" variable, indicating the contribution of each feature in the model's predictions. Several XGBoost models were automatically trained and evaluated using various settings by the AutoML process, which then chose the top model based on the assessment metrics. The provided metrics evaluate the model's effectiveness on the cross-validation and training data folds.

6) *Seasonal Autoregressive Integrated Moving Average (SARIMA)*: SARIMA is an extension of the Autoregressive Integrated Moving Average (ARIMA) model discussed in [3] and [9], which incorporates seasonal and non-seasonal factors in a multiplicative model. Along with the hyperparameters for ARIMA, there are three additional hyperparameters: P is Seasonal autoregressive order; D is Seasonal difference order; q for Seasonal moving average order; and another parameter s, which is the number of time steps for a single seasonal period. A general notation of the SARIMA model is shown as follows:

$$\Phi(B)\Phi_p(B^s)(1-B)^d(1-B^s)^Dy_t = \delta + \theta(B)\Theta_Q(B^s)\epsilon_t$$
where $\Phi(B)$ is ordinary autoregressive, $\theta(B)$ is moving average components with $\Theta_Q(B^s)$ and $\Phi_p(B^s)$ as seasonal autoregressive and moving average components, respectively. $(1-B)^d$ and $(1-B^s)^D$ are the ordinary and seasonal difference components of order d and D, and lastly, ϵ_t is the Gaussian white noise.

- *Seasonal Decompose, Augmented Dickey-Fuller Test, ACF, and PACF plots:* These methodologies are similar to the ARIMA model, so we move to the next step.
- *Parameters with least AIC:* After studying the ACF and PACF plots shown in Fig. [3] and [4], respectively, a range of probable values created combinations that were run through a loop to find the optimal values that gave us the least value for AIC, a combination of (2,1,1)x(0,0,1)12 for parameters. The diagnostics plot for SARIMA is given in Fig. [5]

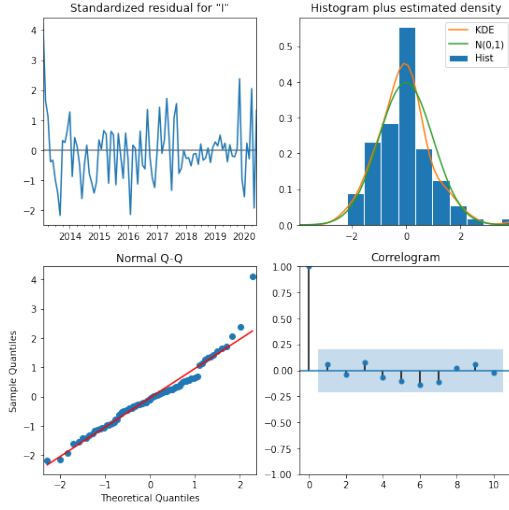


Fig. 5. Diagnostics plot for SARIMA

- *Standardized residuals plot:* There is no obvious pattern in the residuals.
- *Histogram plus estimated density plot:* This shows the distribution of the residuals, with the orange line representing a smoothed version of the histogram and the green line representing the normal distribution.
- *Normal Q-Q plot:* The distribution of most residuals is normal and along the red line except for some in the later half and towards the end.
- *Correlogram plot:* The correlogram plot also shows no significant correlation between the residuals.

7) *Facebook Prophet:* Prophet is an open-source forecasting tool developed by Facebook Inc., and it is available in both Python and R. Prophet was introduced to make forecasting easily tunable and accessible to everyone. The model has a decomposable time series with three components: growth(trend) $g(t)$, seasonality $s(t)$, and holidays $h(t)$ (if applicable). With our inflation dataset, holidays are not applicable. The model can be denoted as: $y(t) = g(t) + s(t) + \epsilon_t$ where ϵ_t represents the variations not taken into consideration by the model and are assumed to be normally distributed. The model treats forecasting as a curve-fitting task; the only regressor is time. Thus, the model is univariate.

- *Renaming time and forecast columns:* The prophet model requires that the date column be renamed to 'ds' and the column forecasted to 'y'.

- *Training:* The most notable part of the prophet model is the low training time. Our dataset only required less than a minute to be trained with about 102 data points.
- *Cross-Validation:* The model was then set up for cross-validation using Prophet's cross-validation method from its library.
- *Finding the best changepoint and seasonality scales:* An iteration through ideal combination values for changepoint and seasonality scales was done to find the best values for which the model would give us the least MAE and MAPE values. Setting the changepoint scale to 0.5 and the seasonality scale to 0.01 would produce the least MAE and MAPE values of 0.239 and 0.041, respectively. So, we trained our model using those parameter values and obtained a model with an RMSE score of 1.294.

IV. RESULTS AND DISCUSSION

A. Data Inspection

Examining the data's graph is the initial stage in time series analysis [12]. The primary purpose of reviewing the data

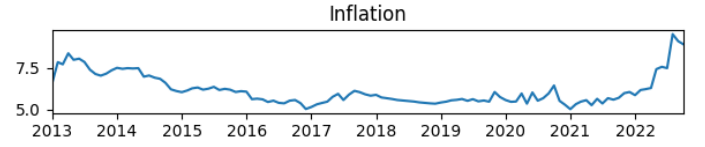


Fig. 6. Inflation plot of Bangladesh

plot is to determine whether there is any discernible pattern or seasonality in the data. Fig. [7] displays Bangladesh's inflation rate from 2013 to 2022. The graph shows no evident seasonality or trend in the data. Also, this graph shows no evidence of steady data. As a result, the graphs in Fig. [7] show that the data lacks seasonality and trend.

B. Checking Seasonality

The Augmented Dicky Fuller (ADF) test can be used to verify the data's stationarity in the second step of developing a time series model. [12]

C. Model Selection

Predicting inflation requires careful consideration when choosing a suitable model. The best model can be chosen using factors like AIC, BIC, and log-likelihood. Lower AIC, BIC, and greater log-likelihood values may indicate the best model.

D. Forecast plots

Here are all the forecasted plots we generated from the different ML models.

E. Evaluation Metrics

Table [1] displays the key findings of this study's objectives. The table displays the relative Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Error (MAE) values for all model settings for all forecast horizons. Additionally, Table 1 provides the relative

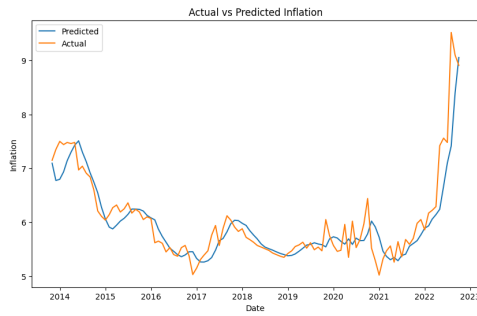


Fig. 7. Inflation forecasted by LSTM

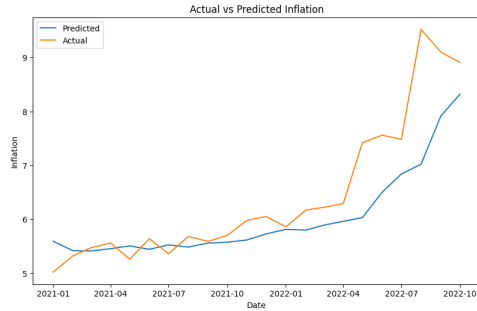


Fig. 8. Inflation forecasted by GRU

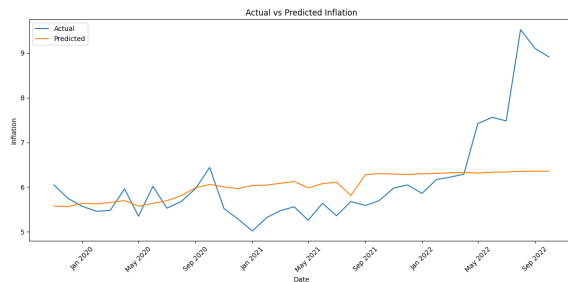


Fig. 9. Inflation forecasted by SVR

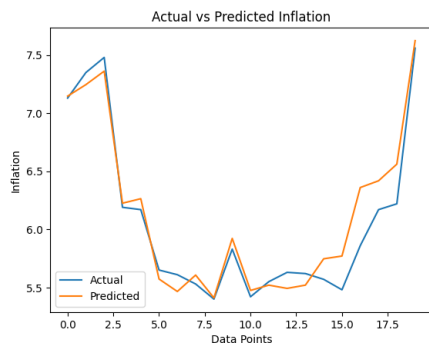


Fig. 10. Inflation forecasted by AutoML

RMSE values, which allow us to make the following key findings from the data:

- Compared to reference models (benchmarks) that employ lags of inflation as predictors, ML approaches can in-

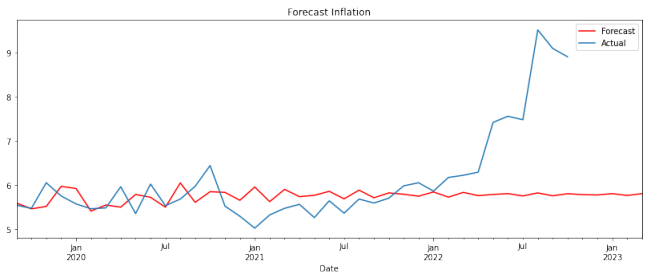


Fig. 11. One-step ahead Inflation forecasted by ARIMA

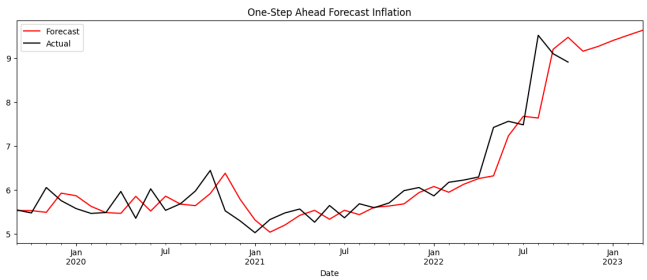


Fig. 12. One-step ahead Inflation Forecasted by SARIMA

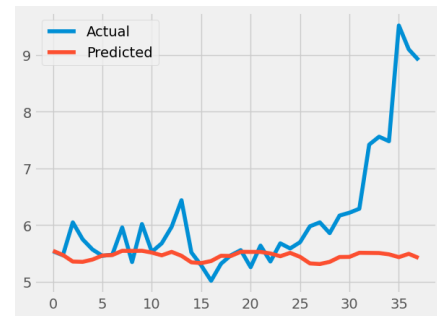


Fig. 13. Inflation forecasted by Prophet

TABLE I
EVALUATION OF EACH MODEL

Evaluation Metrics	LSTM	SVR	GRU	AutoML	ARIMA	SARIMA	Prophet
MAE	0.199	0.646	0.457	0.085	0.623	0.478	0.715
MAPE	3.21%	9.51%	6.82%	3.13%	9.11%	5.17%	9.74%
RMSE	0.313	0.977	0.707	0.122	0.986	0.478	1.249

crease the quality of forecasting Bangladeshi inflation.

- The ensemble methods (LSTM and XGBoost from AutoML) predict average inflation better than the base model from the second month onwards.
- The specification with untransformed data outperformed the other five models when projecting inflations in specific months and average inflation over the entire prediction horizon. A similar conclusion is reached when comparing the outcomes of the LSTM and AutoML models using stationary and non-stationary data.
- The regularized models offer less precise predictions and overall forecasting horizons. Except for forecasts one month in advance, the SARIMA and SVR models

and Prophet produced lower outcomes than the original ARIMA model.

F. Time Performance Metrics

This bar graph visualizes each model's time to train in minutes. The ARIMA, SARIMA, and Prophet models are very time-efficient, taking only 2-3 seconds to train on about 100 data points. The LSTM model takes the most time to train, nearing around 120 minutes. These time values have been accumulated from running the models on the standard Google Colaboratory cloud platform and a code editor across devices with standard 8 GB RAM. Training times differ based on algorithm complexity, and optimization for quick training is essential for real-time, high-frequency forecasting scenarios.

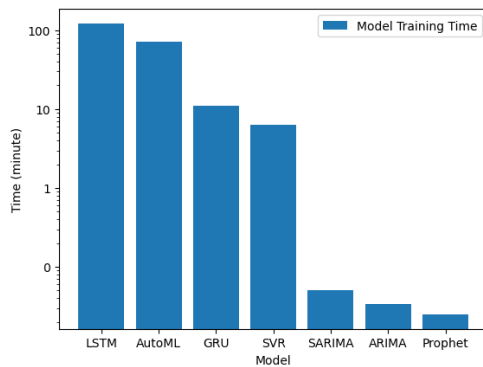


Fig. 14. Training time for each model

G. Limitations and Future Works

Our research on inflation in Bangladesh relied on historical data and presumptively constant relationships between variables due to limited data availability. Accuracy could be increased using updated data, including more economic indicators (interest rate, mortgage rates, etc.), and exploring dynamic modeling approaches. Interdisciplinary collaboration and cross-country analyses could offer broader financial insights for future research on inflation. Based on our findings, policymakers in Bangladesh should consider using LSTM and AutoML models for more accurate inflation forecasting.

V. CONCLUSION

This study's objective was to provide a preliminary assessment of the ability of machine learning techniques to predict inflation in Bangladesh. Two ARIMA variations, a support vector regressor, an extreme gradient boost by AutoML, Prophet, gated recurrent units, and a long short-term memory (LSTM) model, were used to generate the forecasts. Their qualities were assessed per the optimal forecasting literature, and their accuracy was examined with other articles that attempted to forecast inflation. As a result, the analysis of this study suggests utilizing the LSTM model and XGBoost by AutoML to anticipate the rate of inflation using the monthly data for the macro variables that are currently accessible. Based on these results, Bangladesh might benefit from applying machine

learning approaches for predicting inflation. Application of the study's under-performing approaches might be improved as a first step, coupled with the potential expansion of the work to include new ML techniques. Therefore, machine learning methodology and forecast combination approaches are promising alternatives to traditional forecasting tools for producing Bangladesh's short- and medium-term inflation forecasts.

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